The Problem

Biotech equity research and M&A teams price binary clinical and regulatory risk using static, published probability-of-success tables that bucket by phase and broad therapeutic area only. A Phase 2 oncology asset with a rigorous, biomarker-selected trial design and a first-in-class mechanism gets the same default PoS as a Phase 2 oncology asset with a weak, unselected trial, and this leads to very different valuations because it feeds directly into risk-adjusted NPV. It is a plausible contributor to why biotech stocks are notorious for +/-40-50% single-day moves on trial data.

The Data

There are a few core sources of data that we intend to use. The first is the Aggregate Analysis of ClinicalTrials.gov, Duke/CTTI (AACT), which provides a full historical trial registry. We will also use openFDA / Drugs@FDA, which provides approval outcomes and regulatory designations. We will also use the BIO / Informa / Biomedtracker "Clinical Development Success Rates" reports, which provide the published phase-by-therapeutic-area PoS tables that serve as the industry-standard baseline this project tests against. There are also two additional sources we could use, ENCODE and CDC WONDER, for optional enrichment.

The Hypothesis

The primary hypothesis we are testing is that a model trained on features such as mechanism, endpoint type, biomarker-driven patient selection, prior-phase effect size, sponsor track record, and regulatory designations will be better calibrated than the static, phase-only historical table.

The Modeling

Each phase transition is framed as supervised binary classification. We would also use a gradient-tree model as the primary model (XGBoost or LightGBM), and logistic regression as the interpretable baseline. The evaluation uses proper scoring rules: Brier score, log loss, and reliability/calibration curves. It will be benchmarked against the lookup table applied to the same held-out programs. A time-based train/test split (e.g., train on pre-2018 cohorts, test on 2018-2024 outcomes) avoids lookahead bias and mimics how the model would have performed if used in real time.

If time allows, the core AACT/FDA feature set is enriched with two optional signals: an ENCODE-derived target-expression feature and a CDC WONDER-derived disease-burden feature (see The Data).

This work is split across our four-person team: two members lead data acquisition and feature engineering, one leads modeling and evaluation, and one leads the application outputs and final report.

Note: we received AI assistance mainly on this section.

The Output

1. A calibration comparison of the model vs. the industry benchmark against real historical outcomes (reliability diagrams and a Brier-score table).

2. A ranked watchlist of current, still-pending clinical programs where the model's PoS estimate diverges materially from the naive industry benchmark, flagging where the market may be mispriced right now.

3. An illustrative rNPV recalculation for two or three real public biotech assets, showing how the valuation shifts when the model's PoS replaces the benchmark's.

Why This Matters

Risk-adjusted valuation underpins biotech equity research and M&A deal structuring. This is not only for academic interest, it also has practical value. Existing academic PoS-prediction work (the MIT Sloan clinical-trial-outcomes project, the Novartis DREAM challenge, and oncology-specific ML studies in journals like Scientific Reports) builds predictive models but rarely benchmarks directly against the specific static tables practitioners actually use as their default input today. This project closes that particular gap.

The persona of interest would be investment bankers and healthcare-focused equity researchers who rely on these tables for valuation.